SOIL EROSION AND CONSERVATION · NOTE ER07

Runoff Estimation — Rational Method, Empirical Flood Formulae and SCS Curve Number

Course topics: SWCE-502 T6, SWCE-502 T8, SWCE-502 T10, SWCE8.12 L2

Why estimate runoff

- Runoff estimation finds the total volume, peak rate or safe yield of a watershed for a design storm frequency. Direct measurements are scarce, so prediction methods are used to design conservation structures, drainage networks, tanks, ponds, dams, reservoirs, spillways, waterways and outlets.
- Small watersheds need the design peak; large ones the volume or yield. Check estimates against observed data from similar watersheds.
- **Methods:** rational, Curve Number (CN), hydrograph, and empirical formulae (Dickens, Ryve, Inglis).

Rational method (peak runoff)

$$Q = (1/360)\cdot C\cdot I\cdot A = 0.0028\cdot C\cdot I\cdot A$$

Q = peak runoff (m^3/s); C = runoff coefficient (0–1); I = intensity for a duration equal to Tc (mm/hr); A = area (ha)

FPS form: $Q = C\cdot I\cdot A$ (ft^3/s, in/hr, acres)

$$\text{Weighted runoff coefficient: } C = (c_1a_1 + c_2a_2 + \ldots + c_na_n)/(a_1 + a_2 + \ldots + a_n)$$

- C depends on slope, soil, land use and vegetation.
- **Applies** mainly to watersheds ≤ 1300 ha. **Assumes** uniform rainfall intensity lasting at least Tc, spread uniformly over the watershed; longer storms give lower peaks because intensity falls.
- **Limits:** these conditions rarely occur, and tabulated C values may not fit a particular watershed — yet the method is accurate enough for small watersheds, especially where failure has limited consequences.

Illustration (not in the notes)
C = 0.4, I = 60 mm/hr, A = 50 ha
Q = 0.4 × 60 × 50 / 360 = 3.33 m³/s

Empirical flood formulae (large watersheds, monsoon India)

Formula	Equation	Constants and use
Dickens	Q = C·A^(3/4)	C ≈ 11.45–19.42 by rainfall region; moderate river basins of North and Central India
Ryve	Q = C·A^(2/3)	C varies 7.6–10.15
Inglis (fan-shaped catchments)	Q = 123.2·A^0.5	Mainly the Mumbai region
Inglis (small and large catchments)	Q = 123.2·A/√(A + 10.36)	Mainly the Mumbai region

Q = flood discharge (m^3/s); A = catchment area (km^2).

Curve Number (hydrologic soil–cover complex) method

- Developed by the USDA Soil Conservation Service to estimate runoff volume and peak from agricultural watersheds of ≥ 50 ha up to several thousand ha; it can also give the complete hydrograph and water yield.
- It combines soil properties, vegetative cover, land use and management, and physiography — the **soil–cover complex**. Widely used, and adopted with modifications by the Soil Conservation Division, Government of India.

Hydrologic soil groups (HSG)

Soils are grouped by their relative runoff potential and erosion susceptibility using four properties: infiltration rate, hydraulic conductivity, effective depth (readily penetrated by roots) and average clay content of the profile.

Group	Runoff potential	Infiltration (as in the notes)	Typical soils
A	Low	> 25 mm/hr even when wet	Deep, well to excessively drained, gravelly, high conductivity
B	Moderately low	12.5–25 mm/hr	Moderately deep to deep, moderately well to well drained, fine to moderately coarse, moderate conductivity
C	Moderately high	2.5–12.5 mm/hr	Moderately deep to deep, moderately well to well drained, moderately fine to moderately coarse, moderate conductivity
D	High	< 2.5 mm/hr	High-swelling clays, soils with a permanent high water table, clay pan or clay layer near the surface, very shallow soils

USDA TR-55 defines the groups by minimum infiltration of bare soil after prolonged wetting: A > 7.6, B 3.8–7.6, C 1.3–3.8, D < 1.3 mm/h. Quote those values when a question cites the USDA definition.

- **India:** about 53.6% of the land is Group B; the smallest share, 11.09%, is Group A.
- **Combining groups:** one or two groups usually dominate a watershed. Two groups may be merged only if one covers < 3% of the hydrologic unit and they are similar (A–B, B–C, C–D). Example: 90% A + 10% D — the runoff comes mainly from the D area, so treating it all as A under-estimates runoff.

Group (Table 12.4)	Major Indian soils
A	Coastal alluvium; desert soil
B	Alluvial (riverine and deltaic); red and yellow; laterite; tarai; hill and mountain soils; peat; glacier and snow-covered soils
C	Alluvial soil affected by salinity/alkalinity; mixed red and black; forest soil
D	Grey and brown desert soil; black soil (deep, medium, shallow); black soil affected by salinity/alkalinity

Antecedent moisture condition (AMC)

AMC is an index of watershed wetness when rain begins: a dry watershed absorbs more and sheds less. It is the **total rainfall of the 5 days before the storm**, with separate limits for the dormant season (crop dormant or seedlings establishing — lower limits) and the growing season.

AMC	5-day rainfall — dormant	5-day rainfall — growing	Condition
I	< 12 mm	< 36 mm	Dry; less runoff
II	12–28 mm	36–53 mm	Moderately wet; medium runoff
III	> 28 mm	> 53 mm	Saturated; high runoff

> **Example (from the text)**
> Preceding 5-day rainfall: 4.0 + 14.0 + 9.0 + 0.0 + 20.0 = 47 mm
> Growing season: 36 ≤ 47 ≤ 53 → AMC II

Curve number and rainfall–runoff equations

- CN (0–100, dimensionless) expresses runoff potential and depends on land use (cultivated, forest, pasture, wasteland, roads…), cover condition (poor, fair, good) and conservation practice (contouring, terracing…). CN = 100 → impervious, rain ≈ runoff; CN = 0 → all rain infiltrates. Real watersheds lie between.
- Tables give CN for AMC II; convert to AMC I or III with conversion tables, interpolation or graphs.
- **Initial abstraction I_a:** losses before runoff — interception, initial infiltration and depression storage.
- **Puddled transplanted paddy:** first satisfy the standing-water requirement, then estimate runoff from the remaining rain with CN ≈ 95.

$$\text{General: } Q = (P - I_a)^2/[(P - I_a) + S]$$
$$\text{USDA SCS } (I_a = 0.2S)\text{: } Q = (P - 0.2S)^2/(P + 0.8S)$$
$$\text{India, black soils at AMC II and III } (I_a = 0.1S\text{, cracks filled first)}\text{: } Q = (P - 0.1S)^2/(P + 0.9S)$$
$$\text{India, all other cases } (I_a = 0.3S)\text{: } Q = (P - 0.3S)^2/(P + 0.7S)$$
$$CN = 25400/(254 + S) \text{ or } S = 25400/CN - 254 \text{ (S in mm)}$$

Q = direct runoff (mm); P = storm rainfall (mm); S = potential maximum retention (mm).

$$\text{Weighted CN: } CN = (CN_1A_1 + CN_2A_2 + \ldots + CN_nA_n)/(A_1 + A_2 + \ldots + A_n)$$

Correction: the notes omit the bracket around the numerator of the weighted-CN formula; all the products CN_iA_i are summed before dividing by the total area.

- Exclude paddy areas from the weighted CN; use CN ≈ 95 for puddled paddy after deducting its storage. Total runoff = paddy runoff + runoff from the rest of the watershed.

> **Illustration (not in the notes)**
> CN = 75 (AMC II), P = 100 mm, I_a = 0.2S
> S = 25400/75 − 254 = 84.7 mm; I_a = 16.9 mm
> Q = $(100 - 16.9)^2/(100 + 0.8 \times 84.7)$ = 6901/167.7 ≈ 41 mm

Procedure

1. Divide the watershed into uniform hydrologic soil–cover complexes.
2. Find the AMC from the previous 5-day rainfall.
3. Read the CN of each complex from tables.
4. Compute the weighted CN.
5. Convert CN to the prevailing AMC if needed.
6. Find the 6-hour maximum rainfall for the design frequency from records or by interpolation.
7. Compute direct runoff with the appropriate equation.

Peak runoff: $Q_{peak} = 0.020 \cdot A \cdot Q_d / T_p$ (m^3/s; A in ha; Q_d = runoff depth, cm; T_p = time to peak, hr)

Time to peak: $T_p = 0.6 \cdot T_c + \sqrt{T_c}$ (T_c in hr)

Correction: the notes print $T_p = 0.06\ T_c + \sqrt{}\ T_c$. The SCS lag is 0.6 T_c, so the coefficient is 0.6. The 0.020 constant is the SCS triangular-hydrograph value 0.0208 (0.208 with A in km² and Q in mm) rounded.

Sources: ICAR eCourse 'Soil and Water Conservation Engineering' — Dr A. Mishra, reviewed by Dr B. C. Mal (IIT Kharagpur); AgriMoon e-book. USDA-NRCS National Engineering Handbook, Part 630 Hydrology, and TR-55. Further reading: A. M. Michael and T. P. Ojha, Principles of Agricultural Engineering, Vol. II (Jain Brothers).

Practice MCQs

Q1. In Q = CIA/360, A is in

(a) m² (b) ha
(c) km² (d) acres

Q2. The rational method is mainly applicable to watersheds up to about

(a) 50 ha (b) 1300 ha
(c) 10,000 ha (d) 1000 km²

Q3. Dickens' formula is

(a) Q = CA^(2/3) (b) Q = CA^(3/4)
(c) Q = 123.2√A (d) Q = CIA

Q4. In the SCS-CN method with I_a = 0.2S, runoff Q =

(a) $(P - 0.2S)^2/(P + 0.8S)$ (b) $(P - 0.2S)/(P + S)$
(c) $(P - S)^2/P$ (d) P − 0.2S

Q5. For CN = 100, S equals

(a) 0 (b) 25.4 mm
(c) 254 mm (d) infinity

Q6. For black soils at AMC II and III in India, I_a is taken as

(a) 0.1S (b) 0.2S
(c) 0.3S (d) 0.4S

Q7. AMC is determined from rainfall of the previous

(a) 1 day (b) 3 days
(c) 5 days (d) 30 days

Q8. In the growing season, a 5-day rainfall of 60 mm means

(a) AMC I (b) AMC II
(c) AMC III (d) cannot say

Q9. About 53.6% of India's land falls in hydrologic soil group

(a) A (b) B
(c) C (d) D

Q10. Two hydrologic soil groups may be combined if one covers less than

(a) 3% (b) 10%
(c) 25% (d) 50%

Answers

Q1 — (b) ha — *Q in m³/s, I in mm/hr.*
Q2 — (b) 1300 ha
Q3 — (b) Q = CA^(3/4)
Q4 — (a) $(P - 0.2S)^2/(P + 0.8S)$
Q5 — (a) 0 — *Impervious surface.*
Q6 — (a) 0.1S
Q7 — (c) 5 days
Q8 — (c) AMC III — *> 53 mm.*
Q9 — (b) B
Q10 — (a) 3% — *and they are adjacent groups.*